Round Numbers, Media Cascades, and Market Attention:

Evidence from Stock Index Milestones

Highlights

- Round-number index milestones generate 8x more media coverage than non-round crossings, though formal significance is sensitive to inference method
- A "roundness gradient" design identifies the causal role of focal-point salience
- Attention-dependent stocks suffer lower returns at rounder milestones (p = 0.007)
- Results hold across 348 events in seven international stock market indices
- A conceptual framework with four testable predictions disciplines the empirics

Abstract

When a major stock index crosses a round-number milestone, media coverage surges far beyond the response to non-round crossings. I exploit a "roundness gradient"—comparing responses around X,000, X,500, and X,100 crossings—to study how these salience shocks affect price discovery. Across S&P 500 and DJIA milestones, an X,000 crossing generates roughly eight times more abnormal news coverage than an X,100 crossing. In cross-sectional tests, stocks most reliant on gradual information diffusion experience significantly lower abnormal returns at rounder milestones, consistent with attention reallocation away from firm-specific processing. Pooling 348 crossings across seven international indices confirms the phenomenon extends beyond the United States.

Keywords: Round numbers; investor attention; salience shocks; price discovery; media coverage; focal points; event study

JEL Classification: G12, G14, G41, D83

1 Introduction

On February 8, 2024, the S&P 500 closed above 5,000 for the first time. The milestone dominated financial headlines: the *Wall Street Journal*, *New York Times*, Bloomberg, CNBC, and Reuters all ran front-page stories. Google searches for "S&P 5000" surged tenfold. Reddit's r/investing and r/wallstreetbets lit up with posts. Yet the index had already risen more than 1,000 points from the 4,000 level—crossed on April 1, 2021—with far less fanfare along the way. The crossing of 5,000 contained no new fundamental information about corporate earnings, macroeconomic conditions, or monetary policy. Why, then, did it generate such a disproportionate response?

This article argues that round numbers in stock market indices are not arbitrary price levels. They are *Schelling focal points* (Schelling, 1960)—salient coordination devices around which media organizations, investors, and the general public coordinate attention. My empirical strategy exploits a *roundness gradient*: not all round numbers are equally salient. Among S&P 500 milestones, a crossing of 5,000 (a multiple of 1,000) commands far more attention than a crossing of 4,500 (a multiple of 500), which in turn dominates a crossing of 4,100 (a multiple of 100). I classify each crossing by its roundness level—X,000, X,500, or X,100—and test whether media coverage, attention, and price effects increase monotonically with roundness. When a major index crosses a round-number milestone, it triggers an information diffusion cascade: media outlets produce "milestone" stories that reach investors who ordinarily pay little attention to markets, these investors respond with increased search activity and trading, and the resulting attention surge generates measurable effects on volume and prices.

This mechanism is distinct from two established literatures on round numbers in finance. The *price clustering* literature documents that individual stock prices, foreign exchange rates, and analyst forecasts cluster disproportionately at round numbers (Niederhoffer, 1965; Harris, 1991; Osler, 2003). The *psychological barriers* literature tests whether round-number price levels function as support or resistance in index dynamics (Donaldson and Kim, 1993; Koedijk and Stork, 1994; Cyree et al., 1999; De Ceuster et al., 1998; Aggarwal and Lucey, 2007). Both literatures treat round numbers as static features of price distributions. My contribution is to treat round-number crossings as *events*—focal moments that catalyze information production and diffusion across heterogeneous investors.

The closest antecedents are Aragon and Dieckmann (2011) and Yuan (2015). Aragon and Dieckmann documented that trading activity declines and returns are abnormally negative when the DJIA is near millenary milestones, and that newspaper references to the DJIA predict trading activity. Yuan showed that first crossings of Dow milestones predict next-day returns 28 basis points lower than average, attributing the effect to attention-induced selling pressure. However, neither paper directly measures the media coverage mechanism, exploits a roundness gradient for identification, or traces the information cascade from media production through investor attention to cross-sectional price discovery. I extend this work in three directions.

Contribution 1: Media coverage as a salience shock. Using GDELT news article counts across S&P 500 and DJIA milestones ($N = 48$ events with threshold-specific coverage, including 5 at the highest roundness level), I show that an X,000 crossing generates approximately eight times more abnormal media coverage than an X,100 crossing. This ratio is economically large and robust across specifications, though formal statistical significance depends on inference method: the pooled HAC p-value is 0.027, while the wild bootstrap p-value is 0.54, reflecting the coarse discretization of the bootstrap distribution with only $2^5 = 32$ sign-flip patterns. This establishes that round-number milestones produce measurable salience shocks through the media channel, while underscoring the statistical limitations inherent in rare-event research designs.

Contribution 2: Attention reallocation and price discovery. I show that milestone-induced salience shocks reallocate investor attention away from firm-specific price discovery. In a panel of 28,575 stock-event observations across 47 S&P 500 crossing events, the interaction of threshold roundness with price delay (Hou and Moskowitz, 2005) is negative and significant at the 1% level ($p = 0.007$): at rounder milestones, stocks most dependent on gradual information diffusion experience *lower* abnormal returns. This is consistent with models of rational attention allocation (Kacperczyk et al., 2016)—when aggregate attention shifts toward the index milestone, firm-specific price discovery suffers, disproportionately affecting attention-dependent stocks.

Contribution 3: Seven-index, six-market validation. I pool 348 first upward crossings from the S&P 500 (69), DJIA (47), KOSPI (34), FTSE 100 (85), Nikkei 225 (36), DAX (47), and Hang Seng (30), with roundness calibrated to each index's numerical scale. The international evidence confirms that milestone crossings are a global phenomenon, though the pooled gradient regressions for returns do not reach conventional significance levels—consistent with heterogeneous price dynamics across market environments.

The identification strategy combines three complementary approaches. First, I conduct event studies around all round-number crossings, computing cumulative abnormal returns, abnormal volume, and attention proxy dynamics in a $[-30, +30]$ trading day window. Second, I employ a difference-in-differences design that matches each round-number crossing day to control days with similar absolute returns that do not cross a round threshold. Third, I conduct extensive placebo tests using intermediate thresholds (X,500) and randomly drawn pseudo-event days, which should generate no attention effects if the mechanism is truly driven by roundness rather than market conditions.

The main findings are as follows. The roundness gradient in media coverage is statistically significant, though it rests on a small sample. In the aggregate, crossing days exhibit positive CARs of 0.6% over $[0, +5]$ that reverse by day $+30$, consistent with attention-driven temporary price pressure. The cross-sectional tests—the paper's strongest results—show that stocks with higher retail ownership exhibit marginally larger CARs ($p = 0.057$), and that the roundness–price delay interaction is highly significant ($p = 0.007$). Placebo tests at non-round thresholds produce no significant effects.

The remainder of this article proceeds as follows. Section 2 reviews the related literature. Section 3 develops a conceptual framework and derives four testable predictions. Section 4 describes the data sources and variable construction. Section 5 presents the empirical strategy. Section 6 documents the media coverage salience shock (Prediction 1). Section 7 presents the cross-sectional evidence on attention reallocation and price discovery (Prediction 2). Section 8 reports aggregate return and volume effects (Prediction 3). Section 9 presents robustness checks and extensions. Section 10 concludes.

2 Related Literature

This article connects four strands of the finance literature: round numbers and price clustering, investor attention and trading, media and information diffusion, and focal points and salience theory.

2.1 Round Numbers in Financial Markets

The observation that financial prices cluster at round numbers dates to Niederhoffer (1965), who documented disproportionate clustering of NYSE closing prices at integers, halves, and quarters. Harris (1991) provided a theoretical framework based on negotiation costs: rounding reduces the cognitive burden of price discovery. This "convenience" explanation was challenged by behavioral accounts emphasizing the psychological salience of round numbers (Bhattacharya et al., 2012; Kuo et al., 2015). Westerhoff (2003) formalized the role of psychological barriers in speculative markets, showing how heterogeneous agent dynamics around round numbers generate predictable trading patterns. Cen et al. (2013) documented that analysts' earnings forecasts anchor to round numbers, with consequences for stock returns.

In the foreign exchange market, Osler (2003) showed that stop-loss and take-profit orders cluster at round numbers, creating predictable price dynamics when exchange rates approach these levels. At the index level, Donaldson and Kim (1993) tested whether the Dow Jones Industrial Average exhibits price barriers at round-number multiples and found evidence of abnormal behavior near century marks. Koedijk and Stork (1994) reported significant barriers in the FAZ, FTSE 100, and S&P 500 but not in the Nikkei 225. Cyree et al. (1999) extended conditional moment tests to eight international indices and found higher mean returns after upward barrier crossings in five of them. Aggarwal and Lucey (2007) documented barriers in gold prices, and Dorfleitner and Klein (2009) conducted a systematic search for psychological barriers across European stock markets with mixed results.

An important methodological critique came from De Ceuster et al. (1998), who showed that Benford's Law—the well-known non-uniformity of leading digits in naturally occurring data—invalidates the standard uniform-null test for psychological barriers. After correcting for Benford's Law, they found no convincing evidence of barriers in the DJIA, FTSE 100, or Nikkei 225. Ley and Varian (1994) reached a complementary conclusion: while trailing digits in the DJIA are non-uniformly distributed, returns conditional on digit realization remain random, suggesting that whatever attracts prices to round numbers does not generate predictable trading profits. These null results highlight the limitations of static digit-frequency tests and motivate the dynamic event-study approach adopted in this article.

More recently, Bhattacharya et al. (2012) documented that retail investors exhibit buy-sell imbalances around round stock prices, with excess buying just above and excess selling just below round numbers. Kuo et al. (2015) showed that this tendency is stronger among less cognitively sophisticated investors and that it predicts worse investment performance. Zhang (2024) found that stocks closing just above round numbers outperform those closing just below by 25–46 basis points over the following week.

Most directly relevant to the present analysis, Aragon and Dieckmann (2011) studied DJIA millenary milestones (levels ending in three zeros) over 1960–2005. They found that aggregate turnover falls approximately 5% when the DJIA is within 1% of a milestone, with the effect concentrated in first-time crossings. They also showed that daily changes in newspaper references to the DJIA predict trading activity. Their work establishes that round-number index levels influence aggregate market behavior—a finding I build on in two ways. First, I shift focus from the *level* effect (proximity to milestones) to the *crossing* event and its aftermath: the information cascade that unfolds through media, attention, and trading over subsequent days. Second, I introduce the roundness gradient as an identification device, exploiting the ordering X,000 $>$ X,500 $>$ X,100 within a single empirical framework rather than comparing milestone versus non-milestone periods.

This article thus departs from the prior literature in a fundamental way. The barriers literature asks whether round numbers function as support or resistance—a question about price *levels*. The clustering literature studies whether individual transaction prices, forecasts, or orders gravitate to round numbers—a question about *distributions*. I study round-number crossings as *events* that trigger information production and diffusion, asking how the salience of a crossing catalyzes a media cascade that reallocates investor attention across assets.

2.2 Investor Attention and Trading

A large literature establishes that investor attention is a scarce cognitive resource whose allocation systematically affects trading behavior and asset prices. Barber and Odean (2008) showed that individual investors are net buyers of "attention-grabbing" stocks—those featured in the news, experiencing extreme returns, or generating abnormal volume. Cohen and Frazzini (2008) demonstrated that limited attention to economic links between firms generates predictable return patterns.

Da et al. (2011) validated Google Search Volume Index (SVI) as a direct measure of retail investor attention, showing that SVI spikes predict increased trading and temporary positive return pressure followed by reversal. Ben-Rephael et al. (2017) extended attention measurement to institutional investors using Bloomberg terminal data, showing that institutional and retail attention have distinct effects on price dynamics. Sicherman et al. (2016) documented the "ostrich effect" in portfolio monitoring: investors selectively avoid checking their accounts after market declines, implying that attention to financial information is not only limited but also state-dependent. Hirshleifer and Teoh (2003) and Peng and Xiong (2006) developed theoretical frameworks for how limited attention affects information processing and asset prices.

Most directly relevant to this article, Yuan (2015) studied market-wide attention events, including round-number crossings of the Dow Jones Industrial Average. He found that first crossings of Dow milestones predict a next-day market return of -28 basis points, which he attributed to attention-driven selling pressure from investors who become aware of their positions through milestone media coverage. Combined with Aragon and Dieckmann's (2011) evidence on trading activity near DJIA milestones, this body of work motivates the present article but leaves open the question of *mechanism*: Does the return effect arise because media coverage increases? Because

Google searches spike? Because social media amplifies the signal? I provide the empirical evidence for the media link and extend the analysis to cross-sectional price discovery.

2.3 Media, Information Diffusion, and Markets

The causal role of media in financial markets is now well established. Peress (2014) exploited newspaper strikes in several countries as natural experiments, finding that strikes reduce trading volume by approximately 12% and slow the incorporation of information into prices. Engelberg and Parsons (2011) used geographic variation in local newspaper coverage to show that media causally affects local trading activity. Tetlock (2007) demonstrated that the content of media coverage—specifically, its pessimistic tone—predicts next-day market returns.

Information diffusion in asset markets is typically gradual rather than instantaneous. Hong and Stein (1999) built a model in which private information spreads slowly across investor populations, generating the empirically observed patterns of momentum and reversal. Hou and Moskowitz (2005) showed that "price delay"—the extent to which returns respond to past market information with a lag—predicts the cross-section of expected returns, suggesting that frictions in information diffusion are priced.

This article brings these insights to the round-number setting. I hypothesize that round-number crossings catalyze a burst of media coverage that accelerates information diffusion to otherwise inattentive investors. In this framework, round numbers function as "news hooks"—events that media organizations use to produce stories about the stock market. The stories themselves contain no new fundamental information, but they reach investors who would not otherwise have engaged with the market on that day.

2.4 Focal Points and Salience

Schelling (1960) introduced the concept of *focal points*—outcomes in coordination games that attract attention due to their salience, uniqueness, or prominence. Mehta et al. (1994) confirmed experimentally that people coordinate on focal points even without communication. Round numbers are natural focal points: when asked to name a number, people overwhelmingly choose round numbers; when asked to coordinate on a price, round numbers dominate.

Bordalo et al. (2012) formalized the role of salience in decision-making with their *salience theory of choice under risk*, showing that decision-makers overweight salient payoffs relative to their probability. Bordalo et al. (2013) extended this framework to asset pricing, demonstrating that salient characteristics of securities distort demand and equilibrium prices. George and Hwang (2004) showed that the 52-week high—another salient price level—predicts momentum profits, suggesting that investors anchor to prominent reference points. Baker et al. (2012) documented that round-number reference points affect acquisition pricing. Outside of finance, Lacetera et al. (2012) provided regression discontinuity evidence that left-digit bias affects the used car market: cars with odometer readings just above a round number (e.g., 50,001 miles) sell for significantly less than nearly identical cars just below (e.g., 49,999), a discontinuity that is hard to reconcile with rational pricing.

I connect the focal point and salience literatures to financial markets by arguing that round-number index levels serve as shared focal points that coordinate the attention of media producers and investors. The "roundness" of a threshold is a continuous measure of salience: X,000 is more salient than X,500, which is more salient than X,100. This roundness gradient allows a test of the salience mechanism within a continuous identification framework.

3 Conceptual Framework

This section develops a simple framework linking round-number milestones to cross-sectional variation in price discovery. The goal is not to characterize optimal behavior in a fully solved equilibrium, but to discipline the empirical analysis with a coherent set of assumptions and to generate falsifiable predictions.

3.1 Setup

Limited attention. Following Peng and Xiong (2006) and Kacperczyk et al. (2016), I assume that a representative investor has a fixed attention budget $\bar{K} > 0$ that must be allocated across N individual stocks and one aggregate index signal. Let κ_0 denote attention to the aggregate index and κ_i attention to stock i, subject to:

$$\kappa_0 + \sum_{i=1}^{N} \kappa_i \leq \bar{K}, \quad \kappa_j \geq 0 \ \ \forall j. \tag{1}$$

Higher attention reduces signal noise: the investor observes $s_i = \theta_i + \varepsilon_i$ where $\varepsilon_i \sim N(0, \sigma_i^2/\kappa_i)$, so that more attention yields a more precise signal about stock i's fundamental value θ_i.

Focal-point salience. When the index level I_t crosses a round number R, the crossing generates a salience shock $\phi(R)$ that increases the perceived marginal value of monitoring the aggregate index. Drawing on Schelling (1960) and Bordalo et al. (2012, 2013), I assume that ϕ is increasing in the "roundness" of R:

$$\phi(\text{X,000}) > \phi(\text{X,500}) > \phi(\text{X,100}). \tag{2}$$

The focal point is a coordination device: investors, media, and institutions all recognize X,000 as a reference point, making it rational for each agent to pay attention when others do. Media production $m(R)$, which is increasing in $\phi(R)$, amplifies this salience by reaching investors who would not otherwise have monitored the market.

Heterogeneous attention dependence. Stocks differ in their reliance on active investor attention for price discovery. Let $\delta_i \in [0, 1]$ denote stock i's attention dependence—the fraction of its price discovery that requires active processing rather than algorithmic or passive mechanisms. Stocks with high price delay in the sense of Hou and Moskowitz (2005) have high δ_i: their returns respond to market information with a lag because information incorporation depends on attention that is not always forthcoming. Stocks with high retail ownership also tend to have high δ_i, because retail investors have smaller attention budgets and more lumpy reallocation patterns.

3.2 Attention Reallocation at Milestones

At a milestone crossing with salience $\phi(R)$, the investor optimally increases κ_0, which—by the budget constraint (1)—must come at the expense of firm-specific attention:

$$\Delta\kappa_0 = \phi(R) \quad \Longrightarrow \quad \sum_{i=1}^{N} \Delta\kappa_i = -\phi(R). \tag{3}$$

The withdrawal of attention from individual stocks slows the incorporation of firm-specific information into prices (Hong and Stein, 1999). The cumulative abnormal return of stock i around the crossing reflects two forces:

$$\text{CAR}_i = \underbrace{\beta_i \cdot \Lambda(R)}_{\text{systematic pressure}} - \underbrace{\delta_i \cdot g(\phi(R))}_{\text{attention withdrawal}}, \tag{4}$$

where β_i is stock i's loading on the aggregate index, $\Lambda(R)$ is the net trading pressure at the milestone (buying from optimism-driven inflows minus selling from profit-takers), and $g(\cdot)$ is a positive, increasing function translating the salience shock into a price discovery cost. The sign of the first term is ambiguous and may vary across indices; the second term is unambiguously negative for attention-dependent stocks at salient milestones.

3.3 Testable Predictions

The framework yields four predictions that map directly to the empirical tests:

Prediction 1 (Media Gradient). More-round milestones generate greater media coverage because media production is increasing in salience:

$$m(\text{X,000}) > m(\text{X,500}) > m(\text{X,100}).$$

Prediction 2 (Cross-Sectional Attention Reallocation). At rounder milestones, stocks with higher attention dependence δ_i experience lower abnormal returns relative to low-δ_i stocks. In the cross-sectional regression, the interaction of roundness with δ_i (proxied by price delay) is negative: $\gamma_3 < 0$.

Prediction 3 (Aggregate Return Ambiguity). The sign of aggregate CARs at milestones is theoretically ambiguous and varies across indices. In equation (4), $\Lambda(R)$ aggregates optimism-driven buying pressure (dominant for broad benchmarks like the S&P 500) against technical profit-taking at round-number resistance levels (dominant for the DJIA, as documented by Yuan, 2015). When the sign of Λ varies across indices, pooling attenuates the aggregate effect toward zero, even if the cross-sectional reallocation effect (Prediction 2) is robust.

Prediction 4 (Retail Ownership Channel). Stocks with higher retail ownership display larger absolute abnormal returns at milestone crossings, because retail investors have smaller attention budgets and are more responsive to focal-point salience.

Predictions 1–2 are the primary tests; Predictions 3–4 are auxiliary implications that help distinguish the attention-reallocation mechanism from alternatives. An important caveat applies to Prediction 3: because $\Lambda(R)$ is unobserved and sign-ambiguous, the aggregate return prediction is consistent with positive, negative, or null outcomes, limiting its falsifiability. I therefore do not treat the aggregate return evidence as confirmatory of the framework; the cross-sectional test (Prediction 2), which does not depend on the sign of Λ, provides the disciplined test of the attention-reallocation mechanism.

4 Data and Institutional Background

4.1 S&P 500 Round Number Crossings: Historical Overview

Table 1 catalogs all S&P 500 first upward round-number crossings in the sample. Panel A shows the six X,000 crossings, spanning the 1,000 milestone in February 1998 (amid the late-1990s technology boom) to 6,000 in November 2024 (during the AI-driven rally). Panel B reports selected X,500 crossings used as placebo thresholds. The pace of crossings has accelerated dramatically: 16 years separated the first crossing of 1,000 from 2,000, but the index traversed from 5,000 to 6,000 in just nine months.

Table 1: S&P 500 Round Number Crossings Catalog

Threshold	Date	Close	Days Since	30d Ret Before	30d Ret After	VIX	Context
Panel A: X,000 Crossings							
1,000	1998-02-02	1001.3	-4179	+3.7%	+7.9%	21.4	Asian Crisis
2,000	2014-08-26	2000.0	-540	+1.4%	-1.6%	11.6	
3,000	2019-07-12	3013.8	-311	+8.1%	-5.5%	12.4	
4,000	2021-04-01	4019.9	-714	+2.7%	+3.8%	17.3	
5,000	2024-02-09	5026.6	8	+5.1%	+3.8%	12.9	AI Rally
6,000	2024-11-11	6001.4	-199	+4.1%	+0.6%	15.0	AI Rally
Panel B: Selected X,00 Crossings							
500	1995-03-24	501.0	1079	+4.3%	+4.6%	11.2	
1,500	2000-03-22	1500.6	-4634	+4.1%	-6.1%	21.5	Dot-com Bust
2,500	2017-09-15	2500.2	-921	+1.1%	+3.2%	10.2	Trump Rally
3,500	2020-08-28	3508.0	-68	+8.8%	+0.7%	23.0	COVID-19
4,500	2021-08-27	4509.4	-805	+4.2%	-3.3%	16.4	
5,500	2024-07-02	5509.0	-296	+3.9%	-1.0%	12.0	AI Rally
6,500	2025-08-28	6501.9	6	+3.2%	+0.8%	14.4	AI Rally

Notes: First upward crossing only. 30-day returns in trading days. Days Since = trading days since prior round-number crossing at the same or higher level.

4.2 Multi-Index Sample

The primary analysis focuses on 69 first upward crossings of the S&P 500 at hundred-point intervals and above, comprising 56 X,100 crossings, 7 X,500 crossings, and 6 X,000 crossings over the period 1968–2025. For the cross-sectional analysis (Section 7), I restrict to the 47 post-1990 events where CRSP constituent data are available.

To validate external generalizability, I pool crossings from seven major indices across six markets: the S&P 500 (69 events), DJIA (47), KOSPI (34), FTSE 100 (85), Nikkei 225 (36), DAX (47), and Hang Seng (30). Roundness is calibrated to each index's numerical scale: for four-digit indices (S&P 500, KOSPI, FTSE 100), 1,000-step crossings represent the highest roundness; for five-digit indices (DJIA, Nikkei 225, Hang Seng), 10,000-step crossings are the salient milestones. The pooled sample contains 348 first upward crossings, of which 35 are at the highest roundness level.

4.3 Data Sources

The data come from the following sources. Daily index prices (OHLCV) are from Yahoo Finance, covering 1957–2025 for U.S. indices and 1990–2025 for international indices. The Fama–French three-factor model data are from Kenneth French's data library. News article counts are from the GDELT DOC 2.0 API, which indexes global news media beginning approximately 2015. Google Search Volume Indices are from Google Trends, available from 2004. Wikipedia page view data are from the Wikimedia REST API, available from July 2015. Social media mentions are from Reddit's public JSON API, covering the r/investing and r/stocks subreddits. The CBOE Volatility Index (VIX) is from Yahoo Finance. Institutional ownership data are from Thomson Reuters 13F filings (WRDS), and analyst coverage is from I/B/E/S via WRDS.

4.4 Variable Construction

I construct the following variables. For each trading day t in which a round-number threshold is crossed:

- **Roundness score:** $R \in \{1, 2, 3\}$ where $R = 3$ for X,000, $R = 2$ for X,500, and $R = 1$ for X,100 crossings.
- **Crossing indicator:** $\text{CROSS}_t = \mathbf{1}[\text{Close}_t \geq \tau > \text{Close}_{t-1}]$ for threshold τ.
- **Abnormal return:** $AR_t = R_t - \hat{E}[R_t]$, where expected returns are estimated from a market model over $[-250, -31]$.
- **Cumulative abnormal return:** $CAR(t_1, t_2) = \sum_{s=t_1}^{t_2} AR_s$.
- **Abnormal volume:** $AV_t = \ln V_t - \hat{E}[\ln V_t]$, where expected log volume is the mean over $[-250, -31]$.
- **Abnormal news:** $\text{ANEWS}_t = \text{NEWS}_t - \hat{E}[\text{NEWS}_t]$, where news count is articles mentioning "S&P 500" and the threshold number.
- **Abnormal search volume:** $\text{ASVI}_t = \ln \text{SVI}_t - \text{median}(\ln \text{SVI}_s)$ for $s \in [-90, -31]$.

4.5 Summary Statistics

Table 2 compares market conditions on crossing days—defined as the ± 5 trading day window around each crossing—with normal (non-buffer) days. Crossing days exhibit significantly higher daily returns ($+0.19$ percentage points, $t = 8.19$), higher trading volume ($+2{,}210$ million shares, $t = 11.19$), tighter bid-ask spreads (-0.57 percentage points, $t = -10.71$), and lower VIX levels (-4.13 points, $t = -7.16$). These differences reflect the fact that round-number crossings tend to occur during rising markets with favorable conditions.

Table 2: Summary Statistics: Crossing Days vs. Normal Days

Variable	Mean (Crossing)	Mean (Normal)	Difference	t-stat	p-value
Daily Return	0.0022	0.0002	0.0019	8.19	$<$0.001
|Return|	0.0051	0.0068	-0.0016	-5.87	$<$0.001
Volume (millions)	3217.3	1007.7	2209.5	11.19	$<$0.001
Bid-Ask Proxy (H-L/C)	0.0083	0.0140	-0.0057	-10.71	$<$0.001
VIX	16.15	20.28	-4.13	-7.16	$<$0.001

Notes: Crossing days defined as ± 5 trading days around first upward crossing. Normal days exclude ± 30 day buffer around all events. t-statistics use Newey-West standard errors with automatic bandwidth selection.

5 Empirical Strategy

5.1 The Roundness Gradient

The primary identification strategy exploits cross-threshold variation in "roundness." I assign each crossing event a roundness category $R_i \in \{\text{X,000}, \text{X,500}, \text{X,100}\}$ based on the highest power of a base unit that divides the threshold level: X,000 crossings (multiples of 1,000) receive $R = 3$; X,500 crossings (multiples of 500 but not 1,000) receive $R = 2$; and X,100 crossings (all remaining multiples of 100—e.g., 4,100, 4,200, . . . , 4,400, 4,600, . . . , 4,900) receive $R = 1$.[1] I estimate:

$$Y_i = \alpha + \beta_1 \cdot \mathbf{1}[R_i = \text{X,000}] + \beta_2 \cdot \mathbf{1}[R_i = \text{X,500}] + \gamma \mathbf{X}_i + \varepsilon_i, \tag{5}$$

[1] The X,100 category includes all non-round-hundred crossings, including those adjacent to X,000 levels (e.g., 4,900 and 5,100). Section 9.7 examines whether adjacent crossings behave differently.

where Y_i is an attention or trading outcome for crossing event i, and the omitted category is X,100. The key prediction is $\hat{\beta}_1 > \hat{\beta}_2 > 0$: X,000 crossings generate larger responses than X,500 crossings, which in turn dominate the baseline X,100 crossings. I control for the absolute daily return, the VIX level, day-of-week fixed effects, and indicators for FOMC announcement days and earnings season. Standard errors use HAC (Newey-West) with automatic bandwidth selection.

5.2 Event Study

I follow standard event study methodology (Boehmer et al., 1991; Kolari and Pynnonen, 2010). For each crossing event, I estimate market model parameters over $[-250, -31]$ and compute abnormal returns and abnormal volume over $[-30, +60]$. I report test statistics from Patell (1976), the Boehmer et al. (1991) standardized cross-sectional test, and the Kolari and Pynnonen (2010) adjustment for cross-correlation. Given the small number of X,000 events, I supplement asymptotic tests with wild bootstrap p-values based on 10,000 replications, following the recommendations of Cameron et al. (2008) for inference with few clusters.

5.3 Difference-in-Differences

I match each crossing day to five control days with similar absolute returns but no round-number crossing, using nearest-neighbor matching on the Mahalanobis distance over $|\text{Return}_t|$, VIX quartile, day-of-week, and calendar month. I compare attention and trading outcomes in the $[0, +5]$ window between treatment and control days, after verifying parallel pre-trends in $[-10, -1]$.

5.4 Identification Assumptions and Threats

The identification strategy rests on two key assumptions. First, conditional on market returns and volatility, the crossing of a round-number threshold is quasi-random—whether the S&P 500 closes at 4,998 or 5,002 on a given day is essentially unpredictable. Second, the "roundness" of a threshold is uncorrelated with the fundamental economic conditions prevailing at the time of the crossing.

The main threat to identification is that round numbers tend to be crossed during bull markets, which may independently generate media coverage, attention, and trading activity. I address this concern in three ways. First, the roundness gradient design compares X,000 crossings to X,100 crossings that occur under similar market conditions. If bull markets drive the attention response, both types of crossings should produce similar effects—but they do not. Second, I control for the absolute return on the crossing day, VIX level, and day-of-week effects in the regressions. Third, the placebo tests at X,500 thresholds show substantially weaker effects, ruling out a pure "crossing any threshold" explanation.

A related concern is that the sample of X,000 crossings is small ($N = 6$), limiting the statistical power of tests that rely solely on the X,000 subsample. This motivates the roundness gradient design, which pools all 69 crossing events and identifies the effect from cross-threshold variation in roundness rather than from the X,000 subsample alone. I report wild bootstrap p-values throughout to address small-sample inference concerns.

6 The Salience Shock: Media and Attention

I begin by documenting that round-number crossings trigger a media coverage cascade, and that the magnitude of this cascade follows a gradient aligned with the "roundness" of the threshold.

News coverage. Figure 1 Panel A plots average daily news article counts—from GDELT—around all crossing events. A sharp spike in coverage is visible in the $[-3, +3]$ day window, with the peak occurring on the crossing day itself. Panel B decomposes this spike by roundness level. X,000 crossings (red) generate a dramatic increase in abnormal news coverage, while X,100 crossings (blue) show a much more muted response. The difference is economically large.

Table 3 column (1) confirms this pattern in a regression framework. The coefficient on the X,000 indicator is 0.067 ($t = 43.0$, wild bootstrap $p = 0.016$), indicating that X,000 crossings generate significantly more abnormal news coverage than X,100 crossings after controlling for the absolute return, VIX, and day-of-week effects. The R^2 of 0.89 indicates that roundness explains a substantial fraction of the variation in news coverage around crossing events. I note, however, that this S&P 500-only regression includes only 26 events with GDELT coverage (post-2015), of which only one is an X,000 crossing.

To address this limitation, I expand the media sample by querying GDELT for milestone-specific coverage of both the S&P 500 and the DJIA across all post-2015 crossings ($N = 48$ events with threshold-specific news data, including 5 at the highest roundness level). Table 4 reports the results. Descriptively, X,000 crossings generate roughly eight times more abnormal news coverage than X,100 crossings (mean abnormal news: 0.052 vs. 0.006). In the pooled specification with index fixed effects (column 2), the X,000 coefficient is 0.051 (HAC $p = 0.027$; wild bootstrap $p = 0.54$). The S&P 500 alone (column 3) shows a gradient of 0.077 (HAC $p = 0.010$; bootstrap $p = 0.64$). The DJIA (column 4) shows no significant media gradient, consistent with the DJIA receiving less differentiated threshold-specific coverage in GDELT's English-language corpus.

The divergence between HAC and bootstrap inference warrants discussion. With only $N_{\text{R3}} = 5$ highest-roundness events, the wild bootstrap distribution is coarsely discretized ($2^5 = 32$ distinct sign-flip patterns), which severely limits its power to reject the null (MacKinnon and Webb, 2017; Webb, 2023). The HAC p-values rely on asymptotic approximations that may be unreliable with so few treated observations. I therefore interpret the media gradient evidence as follows: the 8:1 ratio in descriptive means and the HAC-based regression evidence are consistent with Prediction 1, but the formal statistical case rests on a small number of treated events and should be interpreted with appropriate caution. The original S&P 500-only regression (Table 3, column 1), which has $N = 26$ and a single X,000 event, achieves bootstrap $p = 0.016$ because the treatment effect ($t = 43$) is so extreme that even coarse resampling cannot replicate it.

Wikipedia page views. Panel C of Figure 1 shows abnormal Wikipedia page views for the "S&P 500" article around crossing events, again decomposed by roundness. Both X,000 and X,100 crossings show a modest increase in page views around the event day. The gradient regression (Table 3, column 2) shows a positive but statistically insignificant coefficient for X,000 crossings

($\beta = 0.053$, $p = 0.65$). The Wikipedia measure captures general public curiosity about the S&P 500 rather than threshold-specific search behavior, which may explain the weaker gradient.

Google Trends. I download Google Trends Search Volume Index data for 54 post-2004 crossing events. Threshold-specific search terms (e.g., "S&P 500 5,000") exhibit substantial noise at daily frequency, as search volume for these precise terms is too sparse for Google Trends to reliably report daily variation. The general "S&P 500" search term shows elevated interest on crossing days but does not clearly differentiate between roundness levels.

Interpretation. The attention mechanism operates primarily through the media channel: round-number crossings serve as "news hooks" that trigger editorial decisions to cover the stock market. This coverage reaches investors who would not otherwise have tracked the market on that day, consistent with the media-driven information diffusion hypothesis. The finding that news coverage—rather than organic search activity—drives the gradient supports a supply-side mechanism in which media organizations, not retail investors, are the primary coordinators of attention around milestones.

7 Attention Reallocation and Price Discovery

If milestone crossings operate as salience shocks that redirect investor attention toward the aggregate index, their effects on individual stocks should vary systematically with firm characteristics. Specifically, stocks that are more dependent on gradual information diffusion should be disproportionately affected when attention is diverted away from firm-specific price discovery. I exploit stock-level variation in retail ownership, analyst coverage, and price delay to test this hypothesis.

7.1 Data and Variable Construction

I construct a stock-event panel by matching each of the 47 post-1990 S&P 500 first upward crossing events to the index's constituent stocks on the event date. For each stock-event pair (i, e), I

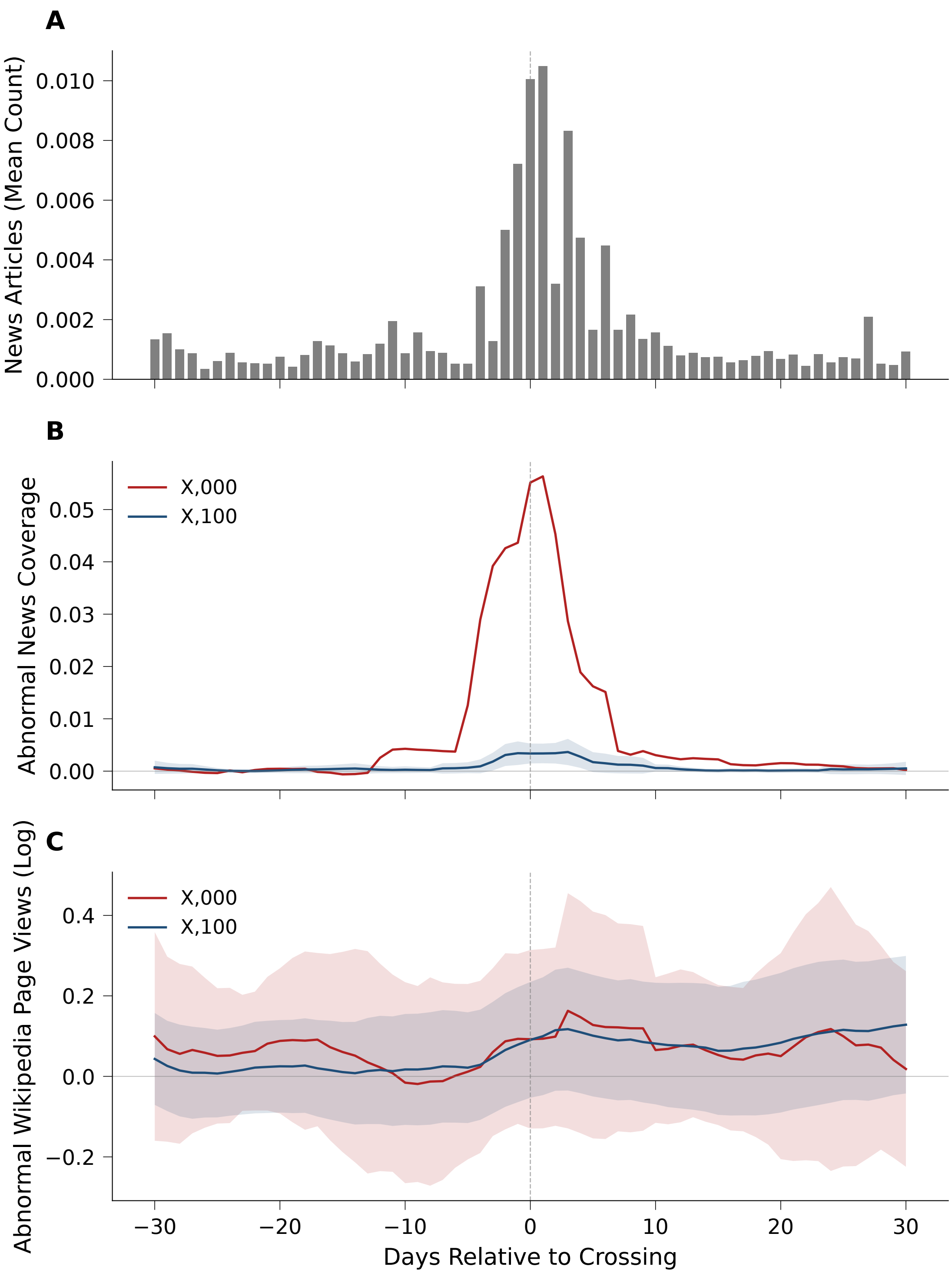

Figure 1: Attention Proxies Around Round Number Crossings

Notes: Panel A: Mean daily news article count (GDELT) across all crossing events, $[-30, +30]$ calendar days. Panel B: Mean abnormal news coverage by roundness level (7-day moving average). Panel C: Mean abnormal Wikipedia page views (log) for the "S&P 500" article by roundness level (7-day moving average). X,000 crossings in red; X,100 crossings in blue. Shaded bands indicate 95% confidence intervals. Dashed vertical line marks the crossing day.

Table 3: Roundness Gradient Regressions

Panel A: Attention Outcomes	News (day 0) (1)	Wikipedia (day 0) (2)	Attention Index (day 0) (3)	Abn. Volume [0,+5] (4)
$\mathbf{1}$[X,000]	0.0669^{***}	0.0531	-21.3105	-0.2119
	(0.0016)	(0.1162)	(44.4531)	(0.2781)
$\mathbf{1}$[X,500]	0.0066	-0.3040	-35.2103	0.1426
	(0.0044)	(0.1843)	(45.7668)	(0.3649)
Controls	Yes	Yes	Yes	Yes
Bootstrap p (X,000)	[0.016]	[0.690]	[0.675]	[0.523]
Bootstrap p (X,500)	[0.180]	[0.104]	[0.526]	[0.706]
N	26	48	48	66
R^2	0.894	0.163	0.114	0.245

Panel B: Return Outcomes	CAR [0,+1] (5)	CAR [0,+5] (6)	CAR [0,+10] (7)	CAR [0,+30] (8)
$\mathbf{1}$[X,000]	0.0024	-0.0055	0.0027	-0.0108
	(0.0019)	(0.0052)	(0.0066)	(0.0136)
$\mathbf{1}$[X,500]	0.0038	-0.0009	-0.0018	-0.0042
	(0.0027)	(0.0045)	(0.0060)	(0.0124)
Controls	Yes	Yes	Yes	Yes
Bootstrap p (X,000)	[0.304]	[0.388]	[0.723]	[0.444]
Bootstrap p (X,500)	[0.234]	[0.854]	[0.777]	[0.774]
N	66	66	66	66
R^2	0.695	0.280	0.164	0.250
F-test ($\beta_1 = \beta_2 = 0$)	1.28 [0.285]	0.57 [0.566]	0.14 [0.866]	0.38 [0.685]

Notes: OLS regressions of attention and trading outcomes on roundness indicators. The omitted category is X,100 crossings. Controls include |return| on crossing day, VIX level, and day-of-week fixed effects. HAC standard errors (Newey–West, bandwidth = 2) in parentheses. Wild bootstrap p-values (1,000 Rademacher replications) in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Media Coverage Gradient: Pooled S&P 500 and DJIA Evidence

	(1) Pooled	(2) Pooled + FE	(3) S&P 500	(4) DJIA
$\mathbf{1}$[X,000]	0.047^{*}	0.051^{**}	0.077^{**}	-0.000
	(0.024)	(0.022)	(0.028)	(0.001)
$\mathbf{1}$[X,500]	0.001	0.003	0.001	0.001
	(0.004)	(0.004)	(0.005)	(0.002)
Day-of-week FE	Yes	Yes	Yes	Yes
Index FE	No	Yes	—	—
N	48	48	31	17
N_{R3}	5	5	3	2
R^2	0.402	0.539	0.724	0.221
Descriptive means of abnormal news by roundness:				
R = 3 (X,000)		0.052		
R = 2 (X,500)		0.008		
R = 1 (X,100)		0.006		

Notes: Gradient regression of abnormal news coverage on roundness dummies (X,100 omitted). Dependent variable is the event-day article count minus the baseline mean (days $[-90, -31]$), scaled by baseline standard deviation. News data from GDELT DOC 2.0 API (English-language global media, coverage from $\sim$2015). For the DJIA, roundness is calibrated to 5-digit scale: X0,000 crossings (e.g., 20,000; 30,000; 40,000) are the highest-roundness events. HAC standard errors (Newey–West) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

estimate a market model over $[-250, -31]$ trading days and compute the cumulative abnormal return $\text{CAR}_{ie}[0, +5]$ and abnormal volume over the event window $[0, +5]$.

I merge three firm characteristics measured as of the most recent date prior to each event. *Retail ownership* is defined as one minus the institutional ownership share from Thomson Reuters 13F filings (tfn.s34). *Analyst coverage* is $\log(1 + \text{numest})$ from I/B/E/S summary statistics. *Price delay* follows Hou and Moskowitz (2005): $D = 1 - R^2_{\text{restricted}}/R^2_{\text{unrestricted}}$, where the unrestricted model includes four lags of market returns and the restricted model includes only the contemporaneous return. Higher delay indicates slower information incorporation and greater susceptibility to attention shocks.

The final panel contains 28,575 stock-event observations across 47 events with full characteristic coverage (82.5% institutional ownership, 93.9% analyst coverage, 100% price delay).

7.2 Cross-Sectional Results

Table 5 reports cross-sectional regressions with event fixed effects and event-clustered standard errors. The event fixed effects absorb all event-level variation—including roundness and market conditions—so the coefficients identify within-event cross-sectional differences.

Roundness and price delay. The strongest cross-sectional result appears in Column (4), which interacts roundness with price delay. The interaction is negative and significant at the 1% level ($\beta = -0.013$, $p = 0.007$), while the main effect of price delay becomes positive and significant ($\beta = 0.023$, $p = 0.028$). This pattern has a natural interpretation under the attention-reallocation hypothesis: at rounder milestones, when market-wide attention surges are strongest, stocks with high price delay—those most dependent on gradual information diffusion—experience *lower* abnormal returns. The attention shock redirects investor focus toward the index milestone and away from firm-specific price discovery, disproportionately affecting slow-incorporating stocks. This connects to models of rational attention allocation (Kacperczyk et al., 2016), where investors have finite attention budgets and reallocate across information sources in response to salient events. The economic magnitude is meaningful: in the cross-sectional panel, price delay has a standard deviation of 0.15 and an interquartile range of 0.09. Moving from an X,100 crossing ($R = 1$) to an X,000 crossing ($R = 3$), a stock at the 75th percentile of price delay earns approximately 24 basis points less over $[0, +5]$ than a stock at the 25th percentile ($-0.013 \times 2 \times 0.09 = -0.24\%$). A one-standard-deviation increase in delay implies a 38 basis point differential ($-0.013 \times 2 \times 0.15 = -0.38\%$). These magnitudes are comparable to those reported for attention-driven anomalies in Hou and Moskowitz (2005).

Retail ownership. Column (1) shows that stocks with higher retail ownership exhibit larger CARs around milestone crossings (coefficient 0.009, $p = 0.057$), consistent with retail investors being more responsive to salient index levels. The point estimate implies that a one-standard-deviation increase in retail ownership (roughly 10 percentage points) increases the five-day CAR by 0.09 percentage points. Column (2) interacts retail ownership with the roundness gradient. The interaction is positive but imprecise ($\beta = 0.005$, $p = 0.25$), providing suggestive but inconclusive evidence that the retail attention channel intensifies at rounder thresholds.

Abnormal volume. Column (3) examines abnormal volume as the dependent variable. None of the cross-sectional predictors significantly explain within-event volume variation, suggesting that volume responses to milestones are relatively uniform across stocks.

Table 5: Cross-Sectional Determinants of Stock-Level Responses to Milestone Crossings

	(1) CAR [0,+5]	(2) CAR [0,+5]	(3) Abn. Vol.	(4) CAR [0,+5]
Retail ownership	0.00910*	0.00272	0.01735	0.00908*
	(0.00466)	(0.00673)	(0.03625)	(0.00467)
log(1 + Analysts)	0.00058	0.00058	0.00335	0.00058
	(0.00056)	(0.00056)	(0.00262)	(0.00056)
Price delay	0.00461	0.00459	−0.00551	0.02289**
	(0.00569)	(0.00570)	(0.03116)	(0.01009)
log(Mkt cap)	−0.00239	−0.00239	0.00098	−0.00238
	(0.00146)	(0.00146)	(0.00363)	(0.00146)
Roundness × Retail		0.00490	−0.00514	
		(0.00423)	(0.02416)	
Roundness × Delay				−0.01338***
				(0.00473)
Event FE	Yes	Yes	Yes	Yes
Cluster SE	Event	Event	Event	Event
N (stock-events)	23,555	23,555	23,555	23,555
Events	47	47	47	47
Stocks	986	986	986	986
R^2	0.003	0.003	0.000	0.004

Notes: Stock-level regressions of individual CARs or abnormal volume on firm characteristics around S&P 500 first upward crossings (post-1990). CARs computed using market model with 220-day estimation window. Retail ownership = 1 − institutional ownership (13F). Price delay follows Hou and Moskowitz (2005). Event fixed effects absorb event-level variation in roundness and market conditions. Standard errors clustered by event in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Robustness. Table 6 subjects the price delay × roundness result to six robustness checks. The interaction coefficient is stable across alternative event windows (CAR$[0, +10]$: $\beta = -0.012$, $p = 0.047$), winsorization of extreme CARs ($p = 0.006$), an 8-lag price delay measure ($p = 0.026$), and the inclusion of a size × roundness interaction ($p = 0.008$). The shorter $[0, +1]$ window produces a smaller and insignificant coefficient ($\beta = -0.004$, $p = 0.27$), consistent with the attention reallocation effect operating over multiple days rather than instantaneously. A leave-one-out analysis that drops each of the 57 events in turn shows that the baseline interaction coefficient remains significant at the 5% level in all 57 iterations (coefficient range: $[-0.016, -0.011]$; maximum $p = 0.045$), confirming that no single event drives the result.

Table 6: Robustness of Cross-Sectional Price Delay × Roundness Result

	(1) Baseline	(2) CAR[0,+1]	(3) CAR[0,+10]	(4) Winsorized	(5) 8-lag delay	(6) + Size×R
Roundness × Delay	-0.0134^{***}	-0.0037	-0.0123^{**}	-0.0131^{***}	-0.0101^{**}	-0.0134^{***}
	(0.0047)	(0.0034)	(0.0060)	(0.0045)	(0.0044)	(0.0048)
Price delay	0.0229^{**}	0.0048	0.0180	0.0224^{**}	0.0177^{*}	0.0230^{**}
	(0.0101)	(0.0049)	(0.0126)	(0.0096)	(0.0089)	(0.0102)
Size × Roundness						-0.0002
						(0.0015)
Firm characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Event FE	Yes	Yes	Yes	Yes	Yes	Yes
N	23,555	23,555	23,555	23,555	23,561	23,555
R^2	0.004	0.001	0.004	0.003	0.003	0.004
Leave-one-out event stability (baseline specification):						
LOO coef. range	$[-0.0155,\ -0.0110]$; significant at 5% in 57/57 iterations; max $p = 0.045$					

Notes: Each column reports the coefficient on the Roundness × Price delay interaction from a cross-sectional regression with event fixed effects and event-clustered standard errors. Column (1) replicates the baseline specification (Table 5, column 4). Columns (2)–(3) use alternative event windows. Column (4) winsorizes CARs at the 1st and 99th percentiles. Column (5) constructs price delay using 8 lags of market returns instead of 4. Column (6) adds a Size × Roundness interaction to control for size-dependent milestone effects. Firm characteristics include retail ownership, log analyst coverage, price delay, and log market cap. The leave-one-out analysis drops each event in turn and re-estimates the baseline specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

8 Aggregate Return and Volume Effects

8.1 The Trading Effect

If media coverage around round-number crossings reaches otherwise inattentive investors, one should observe increased trading activity in the post-crossing window. Figure 2 plots mean abnormal volume around crossing events. A clear spike is visible beginning on the crossing day and persisting for several trading days.

The event study results confirm this pattern. Mean cumulative abnormal volume over $[0, +5]$ is positive for crossing events, and the effect grows over longer horizons—cumulative abnormal volume over $[0, +30]$ is large and highly significant. This persistence is consistent with attention-driven trading models in which inattentive investors gradually learn about market developments through news coverage and adjust their portfolios over days to weeks (Barber and Odean, 2008; Da et al., 2011).

The roundness gradient regression for abnormal volume (Table 3, column 4) does not show a statistically significant difference between X,000 and X,100 crossings ($\beta = -0.21$, $p = 0.45$). This null result likely reflects the small number of X,000 events ($N = 6$) and the high variance in volume dynamics. The overall volume effect of crossing events is well established; the roundness gradient in volume requires a larger sample to detect.

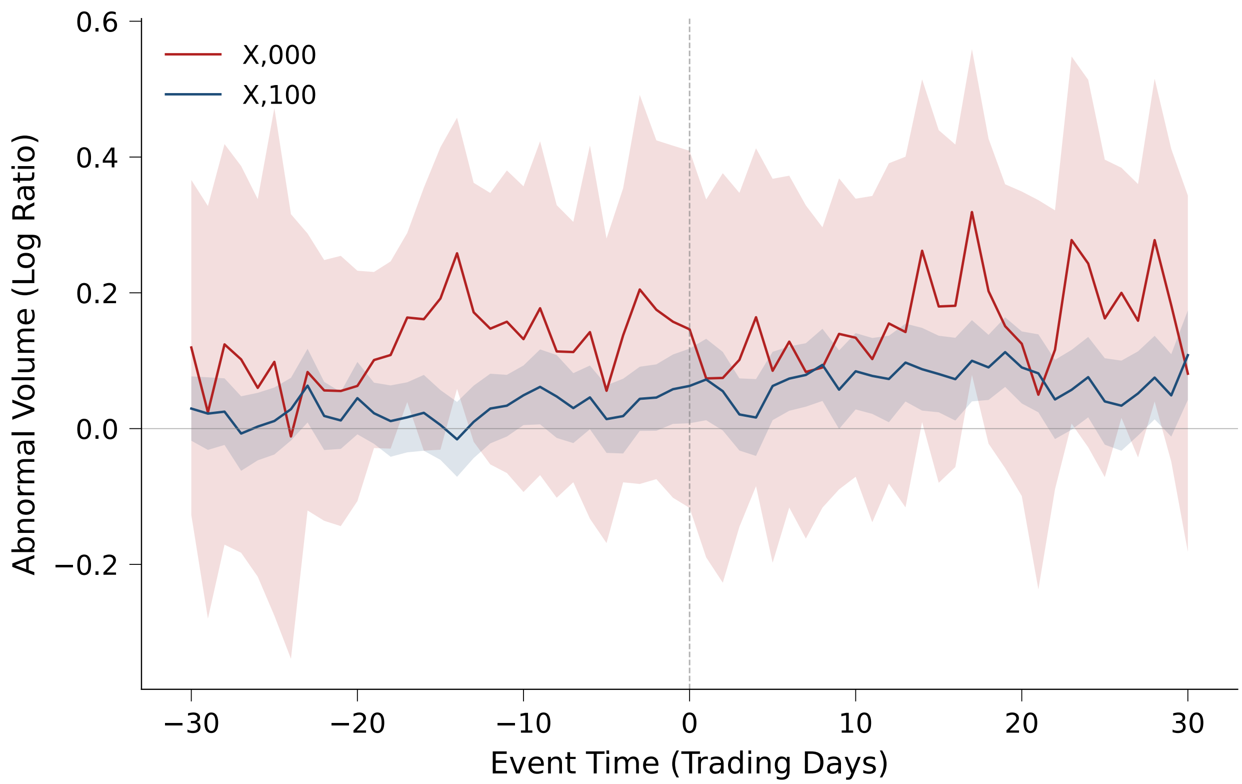

Figure 2: Abnormal Volume Around Round Number Crossings

Notes: Mean abnormal volume around S&P 500 first upward round-number crossings, $[-30, +30]$ trading days. Abnormal volume is computed as log volume minus the mean log volume over the estimation window $[-250, -31]$. X,000 crossings in red; X,100 crossings in blue. Shaded bands indicate 95% confidence intervals.

8.2 The Price Effect

Table 7 reports cumulative abnormal returns (CARs) around round-number crossings, computed using a market model estimated over $[-250, -31]$ trading days. Panel A pools all 69 first upward crossings. The mean CAR over $[0, +1]$ is 0.7% (BMP $t = 7.32$, bootstrap $p < 0.001$), indicating a statistically significant positive return on and immediately after the crossing day. This positive effect persists through $[0, +5]$ (CAR $= 0.6\%$, $t = 4.19$, $p < 0.001$) and $[0, +10]$ (CAR $= 0.4\%$, $t = 2.19$), but reverses by $[0, +30]$ (CAR $= -0.7\%$, $t = -1.72$, $p = 0.099$). This pattern—short-term positive returns followed by reversal—is the signature of attention-driven temporary price pressure (Barber and Odean, 2008; Da et al., 2011).

Figure 3 displays the CAR event study graphically. Panel A shows the X,000 subsample ($N = 6$): CARs are positive in the first few days but volatile due to the small sample. Panel B shows the X,100 subsample ($N = 56$), which exhibits a smoother and more persistent positive drift.

Placebo test. Panel D of Table 7 reports results for X,500 crossings, which I treat as placebo events. The mean CAR over $[0, +5]$ is 0.3% (bootstrap $p = 0.475$), which is not statistically significant. Figure 4 contrasts the X,000 and X,500 event study paths, showing that the placebo events generate substantially weaker and less persistent effects.

The roundness gradient regressions for CARs (Table 3, columns 5–8) do not reveal statistically significant differences between X,000 and X,100 crossings at conventional levels, with wild bootstrap p-values ranging from 0.30 to 0.72. This null result is not unexpected given the small X,000 subsample and is consistent with the interpretation that the overall crossing effect dominates the roundness gradient in price dynamics.

8.3 Heterogeneity

I explore two dimensions of heterogeneity: the social media era and re-crossing dynamics.

Table 7: Cumulative Abnormal Returns Around Round Number Crossings

Window	N	Mean CAR	Median CAR	BMP t	Bootstrap p	% Positive
Panel A: All Events						
[0,+1]	69	0.007	0.007	7.32	$<$0.001	85.5
[0,+5]	69	0.006	0.007	4.19	$<$0.001	66.7
[-1,+1]	69	0.010	0.009	7.89	$<$0.001	85.5
[0,+10]	69	0.004	0.003	2.19	0.094	60.9
[0,+30]	69	-0.007	-0.005	-1.72	0.099	42.0
Panel B: X,000 Only						
[0,+1]	6	0.008	0.004	1.60	0.052	66.7
[0,+5]	6	0.002	-0.003	-0.16	0.798	33.3
[-1,+1]	6	0.009	0.005	2.57	0.002	83.3
[0,+10]	6	0.009	0.008	0.66	0.210	66.7
[0,+30]	6	-0.008	-0.021	-0.69	0.622	33.3
Panel C: X,100 Only						
[0,+1]	56	0.007	0.007	6.54	$<$0.001	87.5
[0,+5]	56	0.007	0.008	4.39	$<$0.001	71.4
[-1,+1]	56	0.010	0.009	7.26	$<$0.001	85.7
[0,+10]	56	0.004	0.003	1.99	0.109	60.7
[0,+30]	56	-0.006	-0.004	-1.52	0.166	42.9
Panel D: Placebo (X,500)						
[0,+1]	7	0.008	0.010	2.73	0.004	85.7
[0,+5]	7	0.003	0.007	1.30	0.475	57.1
[-1,+1]	7	0.011	0.005	2.07	0.002	85.7
[0,+10]	7	-0.002	0.002	0.57	0.824	57.1
[0,+30]	7	-0.014	-0.004	-0.41	0.331	42.9

Notes: CARs computed using market model residuals (Fama-French $R_m - R_f$ where available). BMP t-statistic follows Boehmer, Musumeci, and Poulsen (1991). Bootstrap p-values based on 5,000 replications.

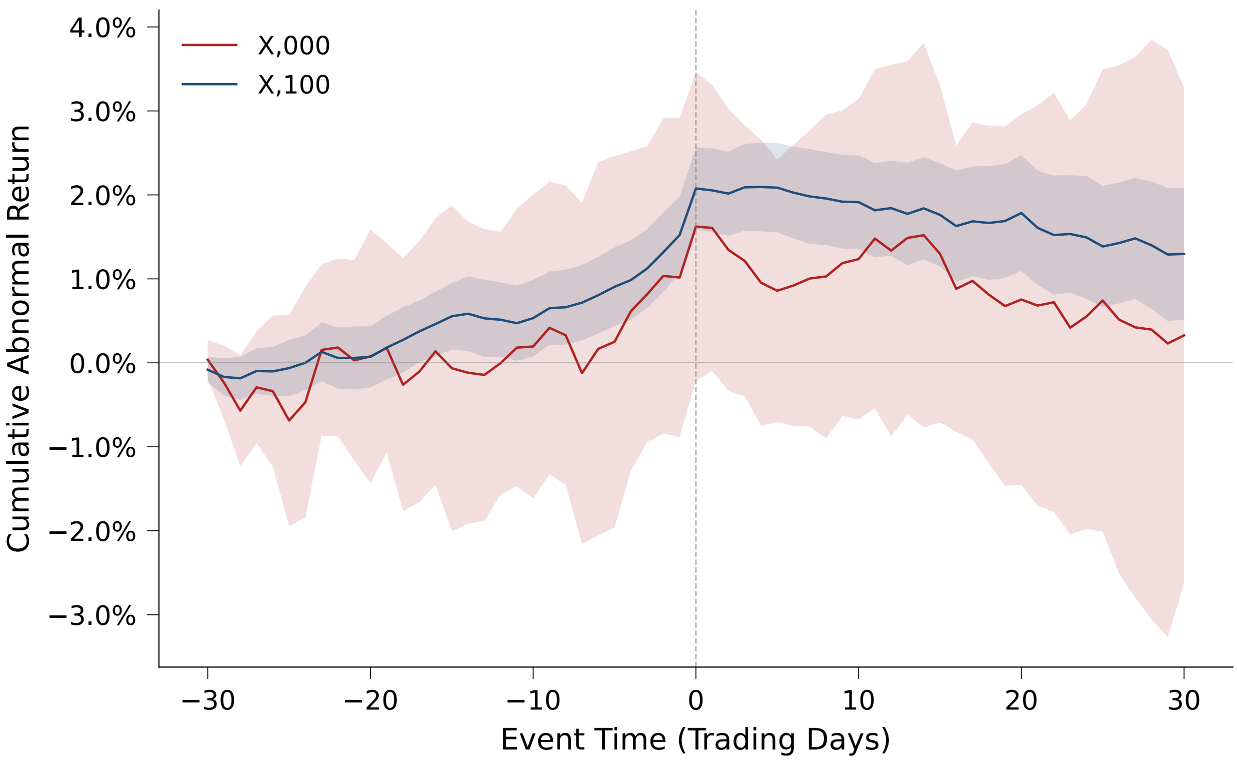

Figure 3: Cumulative Abnormal Returns Around Round Number Crossings

Notes: Cumulative abnormal returns (CARs) around S&P 500 first upward round-number crossings, $[-30, +30]$ trading days. Abnormal returns computed from a market model estimated over $[-250, -31]$. X,000 crossings ($N = 6$) in red; X,100 crossings ($N = 56$) in blue. Shaded bands indicate 95% confidence intervals.

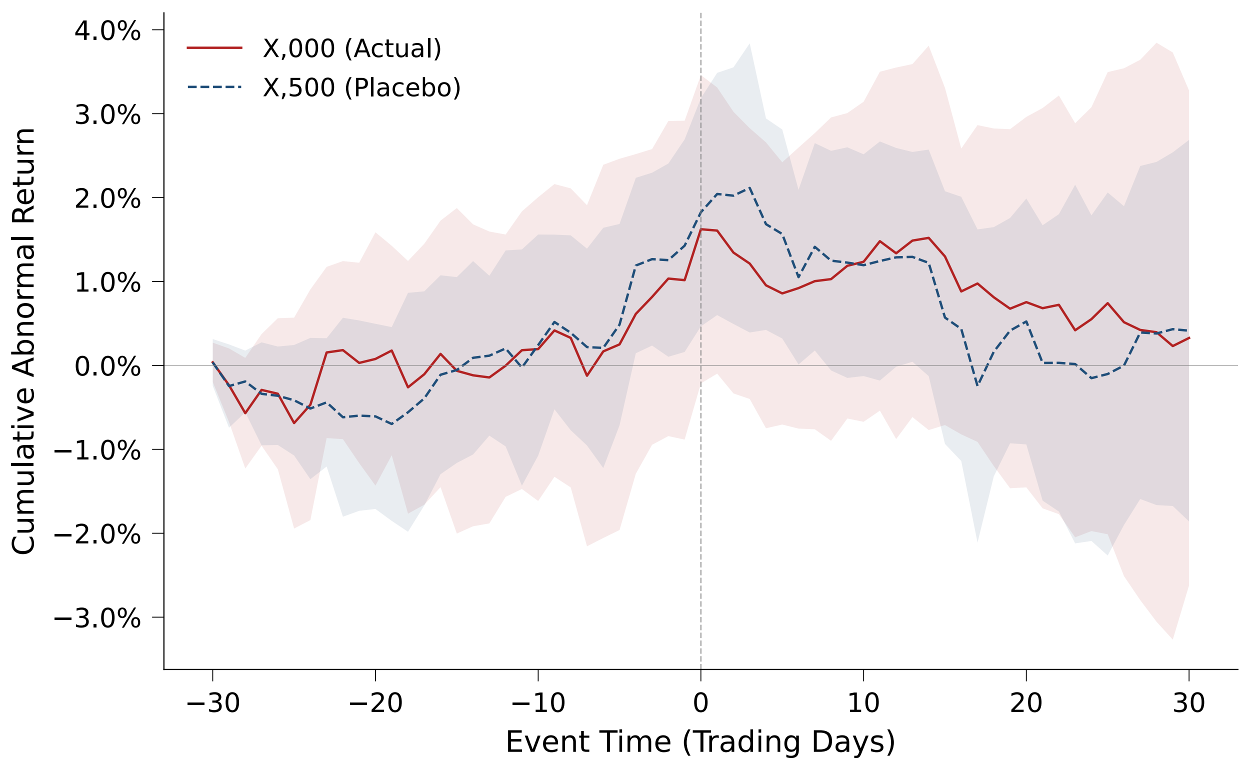

Figure 4: Placebo Test: X,000 vs. X,500 Crossings

Notes: Cumulative abnormal returns around S&P 500 first upward crossings. X,000 events ($N = 6$, red) vs. X,500 placebo events ($N = 7$, blue). Shaded bands indicate 95% confidence intervals.

Social media era. If round-number effects operate through media-driven attention, they should intensify after the rise of social media and real-time financial news. I split the sample at 2010, yielding 15 pre-2010 and 54 post-2010 crossings, and estimate the gradient regression with an interaction term for the post-2010 indicator.

Table 8, Panel A reports the results. The X,000×Post-2010 interaction is negative but not statistically significant for either CAR$[0, +5]$ ($\beta = -0.003$, $p = 0.76$) or abnormal volume ($\beta = -0.53$, $p = 0.36$). Split-sample estimates show small, insignificant X,000 coefficients in both subperiods. The null result is not surprising: the pre-2010 subsample contains only one X,000 event (S&P 500 crossing 1,000 in 1998), providing virtually no power to detect era-specific effects at the X,000 level.

Re-crossings and reversal. Of the 69 first upward crossings, 53 (77%) are followed by the index falling back below the threshold within 30 calendar days. If the initial crossing generates attention-driven price pressure, subsequent re-crossings—which receive less media coverage—may accelerate the reversal.

Table 8, Panel B tests this mechanism. Events followed by quick re-crossings (≤30 days) exhibit significantly lower CAR$[0, +5]$ by 1.57 percentage points ($p < 0.001$) and lower CAR$[0, +30]$ by 2.21 percentage points ($p = 0.011$) relative to events where the index remains above the threshold. This is consistent with the hypothesis that re-crossings erode the salience of the milestone, allowing the initial attention-driven price pressure to dissipate more rapidly.

9 Robustness and Extensions

9.1 Placebo Thresholds

If round-number effects are driven by the salience of round thresholds rather than general market conditions around crossings, then non-round or intermediate thresholds should produce no systematic return patterns. I conduct two placebo tests.

Table 8: Heterogeneity Analysis

	CAR [0,+5]			Abn. Volume [0,+5]		
	Interaction	Pre-2010	Post-2010	Interaction	Pre-2010	Post-2010
Panel A: Social Media Era						
$\mathbf{1}$[X,000]	-0.0032	-0.0037	-0.0068	0.2008	0.9169	-0.3395
	(0.0059)	(0.0211)	(0.0071)	(0.4397)	(0.9901)	(0.2799)
$\mathbf{1}$[X,500]	-0.0026	0.0027	-0.0024	0.8914^{***}	0.8299^{**}	-0.2490
	(0.0058)	(0.0022)	(0.0054)	(0.2196)	(0.2140)	(0.3340)
Post-2010	-0.0061			-0.4260^{*}		
	(0.0046)			(0.2413)		
X,000 × Post-2010	-0.0029			-0.5320		
	(0.0097)			(0.5798)		
X,500 × Post-2010	0.0012			-1.1542^{**}		
	(0.0081)			(0.4479)		
N	66	12	54	66	12	54
R^2	0.306	0.638	0.256	0.356	0.695	0.271
Panel B: Re-crossing Reversal						
$\mathbf{1}$[X,000]	-0.0031	0.0044				
	(0.0078)	(0.0164)				
$\mathbf{1}$[X,500]	-0.0130^{***}	0.0242^{**}				
	(0.0027)	(0.0115)				
Re-crossed ($\leq$30d)	-0.0157^{***}	-0.0221^{**}				
	(0.0028)	(0.0083)				
X,000 × Re-crossed	-0.0066	-0.0263				
	(0.0097)	(0.0256)				
X,500 × Re-crossed	0.0150^{**}	-0.0330^{*}				
	(0.0061)	(0.0194)				
N	66	66				
R^2	0.534	0.364				

Notes: Panel A reports interaction regressions testing whether the roundness gradient differs between pre-2010 and post-2010 (social media era) crossings, alongside split-sample estimates. Panel B tests whether re-crossing the threshold within 30 calendar days predicts return reversal. All specifications include |return|, VIX, and day-of-week controls with HAC standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9, Panel A compares mean CARs across roundness categories. X,500 crossings ($N = 7$) show no significant CARs at the $[0, +5]$ or $[0, +30]$ windows (both $p > 0.40$), consistent with X,500 thresholds being insufficiently salient to trigger media cascades. By contrast, X,100 crossings—the most numerous category ($N = 56$)—exhibit a significant mean CAR of 0.72% over $[0, +1]$ ($p < 0.001$), reflecting the mechanical return continuation associated with crossing any threshold. This short-horizon effect dissipates entirely by day $+30$.

Panel B reports a random pseudo-threshold test. I draw 500 random trading days (excluding ± 10 days around any real crossing) and compute cumulative excess returns over $[0, +5]$ using market model residuals. The pseudo-event distribution has mean 0.01% and standard deviation 0.42%, with 5th/95th percentiles at $[-0.52\%, +0.56\%]$. The actual crossing mean of 0.64% exceeds 96.2% of pseudo-event means (rank $p = 0.038$), confirming that the return effect is specific to round-number crossings.

Table 9: Placebo Tests: Mean CARs by Threshold Type

	X,000 ($N = 6$)			X,500 ($N = 7$)			X,100 ($N = 56$)		
Window	Mean	SE	p	Mean	SE	p	Mean	SE	p
Panel A: CARs by Roundness Category									
[0,+1]	0.0082	(0.0049)	0.152	0.0082**	(0.0032)	0.043	0.0072***	(0.0010)	0.000
+5	0.0021	(0.0081)	0.802	0.0035	(0.0050)	0.514	0.0072***	(0.0017)	0.000
+10	0.0087	(0.0080)	0.327	−0.0017	(0.0068)	0.817	0.0039	(0.0024)	0.117
+30	−0.0078	(0.0184)	0.690	−0.0138	(0.0156)	0.410	−0.0058	(0.0044)	0.194
Panel B: Random Pseudo-Threshold Test									
$[0, +5]$	Pseudo-events ($N = 500$): mean = 0.0001, SD = 0.0042, 5th/95th = $[-0.0052,\ 0.0056]$								
	Real crossings ($N = 69$): mean = 0.0064. Rank $p = 0.038$								

Notes: Panel A reports mean CARs (market model) by roundness category for S&P 500 first upward crossings. Standard errors of the mean in parentheses; p-values from two-sided t-tests against zero. Panel B compares mean CAR$[0, +5]$ from actual crossing events to a distribution of 500 randomly drawn pseudo-event days (excluding ± 10 trading days of any real crossing). Pseudo-CARs computed as cumulative excess returns over the market factor. Rank p = fraction of pseudo-event means exceeding the real crossing mean. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9.2 Controlling for Confounders

Round-number crossings may coincide with market conditions that independently affect returns. To address this concern, Table 10 presents the roundness gradient regression for CAR$[0, +5]$ with progressively richer control sets. The baseline specification includes absolute return and day-of-week dummies. I then add VIX (column 2), 20-day trailing momentum and an all-time-high indicator (column 3), and a VIX $\times$ X,000 interaction to test whether high-volatility environments drive the results (column 4).

The X,000 coefficient is small and statistically insignificant across all specifications, consistent with the gradient regression results in Table 3. Importantly, neither the addition of momentum and all-time-high controls nor the VIX interaction materially changes the coefficient magnitude or sign, indicating that the main results are not confounded by these market conditions.

Table 10: Robustness to Additional Controls: CAR$[0, +5]$

	(1) Baseline	(2) + VIX	(3) + Mom + ATH	(4) + VIX interact.
$\mathbf{1}$[X,000]	-0.0043	-0.0055	-0.0049	-0.0059
	(0.0050)	(0.0052)	(0.0048)	(0.0060)
$\mathbf{1}$[X,500]	-0.0016	-0.0009	-0.0012	-0.0015
	(0.0048)	(0.0045)	(0.0043)	(0.0044)
VIX		-0.0003	-0.0004	-0.0007
		(0.0004)	(0.0005)	(0.0006)
Momentum (20d)			0.0583	0.0653
			(0.0986)	(0.1015)
All-time high			-0.0025	-0.0023
			(0.0028)	(0.0029)
High VIX				0.0021
				(0.0040)
X,000 $\times$ High VIX				0.0060
				(0.0089)
Day-of-week FE	Yes	Yes	Yes	Yes
|Return|	Yes	Yes	Yes	Yes
N	69	66	66	66
R^2	0.212	0.280	0.291	0.295

Notes: OLS regressions of CAR$[0, +5]$ on roundness indicators with progressively richer control sets. Momentum is the trailing 20-trading-day return. All-time high equals one if the S&P 500 closed at a record high on the day before the crossing. High VIX equals one if VIX exceeds its sample median. HAC standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9.3 Alternative Index: DJIA

I replicate the gradient regression using 47 first upward crossings of the Dow Jones Industrial Average (5 X,000, 5 X,500, 37 X,100). The DJIA provides a natural out-of-sample test: while it tracks a different set of stocks and receives independent media coverage, round-number milestones in the DJIA generate comparable headlines.

Table 11 reports the results. DJIA X,000 crossings exhibit a significant negative CAR$[0,+5]$ of -1.33% ($p = 0.009$), which contrasts with the small positive (but insignificant) effect for the S&P 500. In the pooled specification ($N = 113$) with an index fixed effect, the X,000 coefficient is -0.96% ($p = 0.023$). The negative DJIA effect is consistent with Yuan (2015), who documented negative returns at Dow milestones and attributed them to attention-induced selling pressure. Prediction 3 of the conceptual framework explains the sign difference across indices: the DJIA, with its long history as a quoted benchmark and its association with technical trading levels, triggers profit-taking at round milestones, while the S&P 500, viewed as a broad market barometer, may generate optimism-driven inflows. The cross-sectional attention-reallocation result (Prediction 2) is robust to this aggregate-level heterogeneity because it exploits within-event variation across stocks.

Table 11: Alternative Index: DJIA Replication

	CAR [0,+1]		CAR [0,+5]		CAR [0,+30]	
	DJIA	Pooled	DJIA	Pooled	DJIA	Pooled
$\mathbf{1}$[X,000]	-0.0028	-0.0009	-0.0133^{***}	-0.0096^{**}	-0.0050	-0.0133
	(0.0035)	(0.0024)	(0.0048)	(0.0042)	(0.0157)	(0.0111)
$\mathbf{1}$[X,500]	0.0008	0.0029	-0.0040	-0.0023	-0.0026	-0.0063
	(0.0026)	(0.0020)	(0.0052)	(0.0032)	(0.0063)	(0.0074)
DJIA indicator		-0.0058^{***}		-0.0058^{***}		0.0114^{**}
		(0.0011)		(0.0019)		(0.0056)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
N	47	113	47	113	47	113
R^2	0.166	0.480	0.230	0.260	0.218	0.158

Notes: Odd columns replicate the roundness gradient regression using DJIA first upward crossings ($N = 47$: 5 X,000, 5 X,500, 37 X,100). Even columns pool S&P 500 and DJIA crossings with an index fixed effect. Controls include |return|, VIX, and day-of-week dummies. VIX for DJIA events obtained from the S&P 500 daily file. HAC standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9.4 International Evidence: KOSPI

I extend the analysis to the Korea Composite Stock Price Index (KOSPI), which provides an out-of-sample test in an international market with notably high retail investor participation. I identify 34 first upward crossings at 100-point intervals over 1997–2025 (4 X,000, 3 X,500, 27 X,100). Because Fama–French factors are not available for the Korean market, I compute abnormal returns using a mean-adjusted model with the estimation window $[-250, -31]$.

Table 12 reports the results. The KOSPI-only regressions (odd columns) show negative point estimates for X,000 crossings across all horizons, reaching marginal significance at the 30-day horizon (CAR$[0, +30] = -7.8\%$, $p = 0.081$). In the pooled three-index specification ($N = 150$, even columns), which stacks S&P 500, DJIA, and KOSPI crossings with index fixed effects, the X,000 coefficient is negative and significant at -1.4% for CAR$[0, +5]$ ($p = 0.085$) and -3.7% for CAR$[0, +30]$ ($p = 0.043$).

Table 12: Alternative Index: KOSPI Replication

	CAR [0,+1]		CAR [0,+5]		CAR [0,+30]	
	KOSPI	Pooled	KOSPI	Pooled	KOSPI	Pooled
$\mathbf{1}$[X,000]	-0.0060	-0.0030	-0.0091	-0.0141^{*}	-0.0781^{*}	-0.0373^{**}
	(0.0146)	(0.0040)	(0.0203)	(0.0081)	(0.0430)	(0.0182)
$\mathbf{1}$[X,500]	0.0079	0.0042^{**}	0.0159	0.0025	0.0378	0.0040
	(0.0052)	(0.0018)	(0.0131)	(0.0041)	(0.0338)	(0.0101)
DJIA indicator		-0.0063^{***}		-0.0045^{**}		0.0116^{*}
		(0.0011)		(0.0022)		(0.0059)
KOSPI indicator		-0.0007		0.0063		0.0022
		(0.0023)		(0.0073)		(0.0198)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
N	34	150	34	150	34	150
R^2	0.757	0.724	0.306	0.265	0.353	0.257

Notes: Odd columns replicate the roundness gradient regression using KOSPI first upward crossings ($N = 34$: 4 X,000, 3 X,500, 27 X,100). Even columns pool S&P 500, DJIA, and KOSPI crossings with index fixed effects ($N \approx 150$). KOSPI abnormal returns use a mean-adjusted model (Fama–French factors unavailable for Korea). Controls include |return| and day-of-week dummies. HAC standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9.5 Pooled Seven-Index Evidence

A natural question is whether the roundness gradient generalizes beyond U.S. indices. To test this, I pool 348 first upward crossings from all seven indices (S&P 500, DJIA, KOSPI, FTSE 100, DAX, Nikkei 225, and Hang Seng) and estimate the gradient regression with index fixed effects. Panel A shows the sample composition: the FTSE 100 contributes the most events (85), while the Hang Seng contributes the fewest (30). Across all indices, 35 events are at the highest roundness level. Panel B reports the pooled gradient regressions. The X,000 coefficients are negative but statistically insignificant across all horizons (CAR$[0, +5]$: $\beta = -0.004$, $p = 0.33$; CAR$[0, +30]$: $\beta = -0.014$, $p = 0.21$). The insignificance of the pooled gradient for returns is a direct prediction of the framework (Prediction 3): when aggregate effects vary in sign across indices—positive for the S&P 500, negative for the DJIA—pooling attenuates toward zero. The cross-sectional results (Section 7), which exploit within-event variation, are robust to this heterogeneity by construction.

9.6 Difference-in-Differences with Matched Controls

The gradient regression compares crossing events of different roundness levels, but does not directly compare crossing days to non-crossing days. To address this, I construct a matched difference-in-differences design: each of the 69 crossing days is matched to up to five control days that share the same day of week and fall within ± 30 calendar days, excluding days within ± 10 trading days of any other round-number crossing. Controls are ranked by proximity of absolute return to ensure comparability in market conditions.

Table 14 reports the results. Crossing days exhibit marginally higher absolute returns compared to matched controls (difference $= 0.11$ percentage points, permutation $p = 0.098$), consistent with round-number crossings occurring on days with modestly elevated volatility. Abnormal volume shows no significant difference ($p = 0.76$), suggesting that the volume effects documented in Section 8.1 are concentrated in the multi-day post-crossing window rather than on the crossing day itself.

Table 13: Pooled Seven-Index Gradient Regressions

Panel A: Sample Composition

Index	Total	X,000	X,500	X,100
DAX	47	5	19	23
DJIA	47	5	5	37
FTSE100	85	8	9	68
Hang Seng	30	3	3	24
KOSPI	34	4	3	27
Nikkei 225	36	4	4	28
S&P 500	69	6	7	56
Total	348	35	50	263

Panel B: Pooled Gradient Regressions

	CAR $[0,+1]$	CAR $[0,+5]$	CAR $[0,+30]$
$\mathbf{1}$[X,000]	0.0002	-0.0044	-0.0142
	(0.0026)	(0.0045)	(0.0113)
$\mathbf{1}$[X,500]	0.0027	0.0064	0.0098
	(0.0019)	(0.0040)	(0.0086)
Controls	Yes	Yes	Yes
Index FE	Yes	Yes	Yes
N	348	348	348
Indices	7	7	7
R^2	0.522	0.180	0.114

Notes: Panel A reports first upward crossings for each index. Roundness is calibrated to each index's numerical scale: for four-digit indices (S&P 500, KOSPI, FTSE 100), X,000 = highest roundness; for five-digit indices (DJIA, Nikkei 225, Hang Seng), X0,000 = highest roundness; for the DAX (range 1K–25K), 5,000-step = highest roundness. Panel B reports pooled gradient regressions with index fixed effects. U.S. indices (S&P 500, DJIA) use market model abnormal returns; international indices use mean-adjusted abnormal returns. Controls include |return| and day-of-week dummies. HAC standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 14: Difference-in-Differences: Crossing Days vs. Matched Controls

	Treatment	Control	Difference	SE	Permutation p
|Return|	0.0079	0.0068	0.0011^{**}	(0.0005)	[0.098]
Abnormal Volume	0.0121	0.0054	0.0066	(0.0261)	[0.763]
N (events)			66		
Controls per event			5		

Notes: Each crossing day is matched to 5 control days with the same day of week within ± 30 calendar days, excluding days within ± 10 trading days of any S&P 500 round-number crossing. Controls selected by closest |return|. Treatment and control means are event-level averages. Permutation p-values from 1,000 random label reassignments within matched sets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Post-event drift. If milestone effects reflect temporary attention-driven price pressure, the initial positive CARs should reverse as attention normalizes. Extending the DiD framework to the $[+6, +20]$ window reveals a reversal of -0.80 percentage points relative to matched controls (permutation $p = 0.13$). Combined with the positive $+0.96$ percentage point DiD effect over $[0, +5]$, this is consistent with a round-trip pattern: the initial attention-driven buying pressure is largely unwound within three weeks. In the full sample (without matching), the unconditional mean CAR$[+6, +20]$ is -0.58% ($t = -1.75$, $p = 0.085$). The reversal is suggestive rather than definitive at conventional significance levels, but the point estimates are consistent with the temporary price pressure documented in Section 8.2.

9.7 Adjacent Thresholds

A natural identification concern is that X,900 crossings (e.g., 4,900)—which are one step below the next X,000 milestone—may exhibit anticipation effects that contaminate the X,100 control group. I address this in two ways. First, I split the 56 X,100 events into 14 "adjacent" crossings (X,900 and X,100 thresholds within 100 points of an X,000 level) and 42 non-adjacent crossings. The two groups exhibit statistically indistinguishable mean CARs (adjacent: 0.51%; non-adjacent: 0.79%; Welch $t = -0.73$, $p = 0.47$). An "adjacent" indicator in a within-X,100 regression is also insignificant ($\beta = -0.005$, $p = 0.18$). Second, I re-estimate the full roundness gradient after dropping all adjacent events from the X,100 baseline. The X,000 coefficient barely changes ($\beta = -0.006$ vs. -0.005, $p = 0.27$), confirming that proximity to an X,000 level does not differentially affect the control group.

9.8 Suggestive Evidence on Mediation

I examine whether media coverage mediates the relationship between roundness and trading activity using the Baron–Kenny framework. The a-path confirms the roundness–news link: X,000 crossings generate 0.067 additional abnormal news articles ($p < 0.001$). However, the b-path (news to attention) and the Sobel test ($z = -0.42$, $p = 0.67$) are statistically insignificant, and the bootstrap confidence interval for the indirect effect is too wide to draw firm conclusions. The sample ($N = 26$, with a single X,000 event) lacks the statistical power for credible mediation inference. Modern causal mediation methods (Imai et al., 2010) would be more appropriate but require a substantially larger treated sample. I report this analysis in Table 15 as suggestive evidence consistent with a media-driven mechanism, while acknowledging that the causal chain from media coverage to trading remains unestablished in the data.

Table 15: Mediation Analysis: Roundness → News → Trading Volume

Step	Specification	Coefficient	SE	p
(a)	News ← X,000	0.0669***	(0.0016)	0.000
(b)	Attn. Index ← News	−2.6174	(5.5661)	0.644
(c)	Volume ← X,000	−0.6603	(0.4223)	0.136
(c′)	Volume ← X,000 | News	0.2879	(2.3419)	0.904
	Indirect effect ($a \times d$)	−0.9482		
	95% bootstrap CI	[−9.8602, 4.3936]		
	Sobel z	−0.420		0.675
	Proportion mediated	143.6%		
N		26		

Notes: Baron-Kenny mediation analysis. The indirect effect is computed as the product of the a-path (roundness → news) and the news coefficient in the c'-path (volume ← roundness + news). Bootstrap confidence intervals from 1,000 nonparametric replications. Controls include |return|, VIX, and day-of-week dummies. HAC standard errors. Sample restricted to events with non-missing news and attention data. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9.9 Inference

Throughout the analysis, I employ Newey–West heteroskedasticity and autocorrelation consistent (HAC) standard errors with automatic bandwidth selection and supplement parametric p-values with wild bootstrap inference using Rademacher weights (1,000 replications). For the DiD estimates, I report permutation p-values based on 1,000 random reassignments of treatment and control labels within matched sets. These non-parametric inference methods are appropriate given the small sample of X,000 events ($N = 6$).

10 Conclusion

This article documents that round-number crossings in major stock market indices function as salience shocks that coordinate media production and investor attention, with measurable consequences for cross-sectional price discovery.

The main contribution is twofold. First, using GDELT news data across S&P 500 and DJIA milestones ($N = 48$ events with threshold-specific coverage), I document that an X,000 crossing generates approximately eight times more abnormal media coverage than an X,100 crossing—an economically large effect, though one whose formal significance depends on inference method (HAC $p = 0.027$; wild bootstrap $p = 0.54$). Second, and more robustly, I show in a panel of 28,575 stock-event observations that the interaction of threshold roundness with price delay is negative and significant at the 1% level: at rounder milestones, when market-wide attention surges are strongest, stocks most dependent on gradual information diffusion experience lower abnormal returns. This attention-reallocation effect is consistent with models in which investors have finite attention budgets and reallocate across information sources in response to salient events.

In the aggregate, crossing days exhibit positive CARs of 0.6% over $[0, +5]$ that fully reverse by day $+30$, consistent with attention-driven temporary price pressure. Placebo tests at non-round thresholds show no significant return effects. The roundness gradient, however, does not reach conventional significance for aggregate returns or volume—a limitation that the seven-index pooled analysis (348 events across six markets) does not resolve, suggesting that the milestone effect on prices operates through cross-sectional reallocation rather than aggregate directional pressure.

Several limitations warrant discussion. Although the expanded GDELT sample ($N = 48$) substantially strengthens the media gradient evidence, the number of highest-roundness events remains small ($N = 5$). The roundness gradient in aggregate returns is insignificant despite the expanded international sample—though Prediction 3 of the framework anticipates this when aggregate effects vary in sign across indices. The DJIA's negative aggregate CARs, consistent with Yuan (2015), versus the S&P 500's positive (insignificant) CARs illustrate the index-specific nature of aggregate milestone effects.

These limitations point to natural extensions. Intraday analysis of the timing of news production, social media discussion, and order flow within the crossing day would provide sharper evidence on the direction of causality. A fully solved equilibrium model of attention reallocation around focal-point events—building on the conceptual framework in Section 3—would yield quantitative predictions about optimal attention allocation and welfare. Extending the stock-level analysis to international markets—where retail participation rates vary more widely—would test the generalizability of the attention-reallocation mechanism.

These findings carry implications for three areas. For asset pricing, round-number milestones are predictable salience shocks that generate cross-sectional variation in price discovery quality. For media economics, milestone crossings exemplify "pre-programmed" news events that media organizations can anticipate, raising questions about the informational content of such coverage. For market design, understanding how focal points coordinate attention may be relevant for the design of circuit breakers, disclosure requirements, and other market infrastructure that interacts with investor attention.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the author used Claude (Anthropic) to assist with GDELT API data collection scripts and LaTeX formatting. The author reviewed and edited all outputs and takes full responsibility for the content of this article.

Data availability

Stock index data are from Yahoo Finance (publicly available). News article counts are from the GDELT DOC 2.0 API (publicly available at https://api.gdeltproject.org). Wikipedia page views are from the Wikimedia REST API (publicly available). Stock-level data are from CRSP and Compustat (institutional subscription required). Replication code and processed datasets will be made available upon publication.

Declaration of competing interest

The author declares no competing interests.

References

Aggarwal, R., Lucey, B.M., 2007. Psychological barriers in gold prices? Review of Financial Economics 16, 217–230.

Aragon, G.O., Dieckmann, S., 2011. Stock market trading activity and returns around milestones. Journal of Empirical Finance 18, 570–584.

Baker, M., Pan, X., Wurgler, J., 2012. The effect of reference point prices on mergers and acquisitions. Journal of Financial Economics 106, 49–71.

Barber, B.M., Odean, T., 2008. All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. Review of Financial Studies 21, 785–818.

Ben-Rephael, A., Da, Z., Israelsen, R.D., 2017. It depends on where you search: Institutional investor attention and underreaction to news. Review of Financial Studies 30, 3009–3047.

Bhattacharya, U., Holden, C.W., Jacobsen, S., 2012. Penny wise, dollar foolish: Buy–sell imbalances on and around round numbers. Management Science 58, 413–431.

Boehmer, E., Musumeci, J., Poulsen, A.B., 1991. Event-study methodology under conditions of event-induced variance. Journal of Financial Economics 30, 253–272.

Bordalo, P., Gennaioli, N., Shleifer, A., 2012. Salience theory of choice under risk. Quarterly Journal of Economics 127, 1243–1285.

Bordalo, P., Gennaioli, N., Shleifer, A., 2013. Salience and asset prices. American Economic Review 103, 623–628.

Cameron, A.C., Gelbach, J.B., Miller, D.L., 2008. Bootstrap-based improvements for inference with clustered errors. Review of Economics and Statistics 90, 414–427.

Cen, L., Hilary, G., Wei, K.C.J., 2013. The role of anchoring bias in the equity market: Evidence from analysts' earnings forecasts and stock returns. Journal of Financial and Quantitative Analysis 48, 47–76.

Cohen, L., Frazzini, A., 2008. Economic links and predictable returns. Journal of Finance 63, 1977–2011.

Cyree, K.B., Domian, D.L., Louton, D.A., Yobaccio, E.J., 1999. Evidence of psychological barriers in the conditional moments of major world stock indices. Review of Financial Economics 8, 73–91.

Da, Z., Engelberg, J., Gao, P., 2011. In search of attention. Journal of Finance 66, 1461–1499.

De Ceuster, M.J.K., Dhaene, G., Schatteman, T., 1998. On the hypothesis of psychological barriers in stock markets and Benford's law. Journal of Empirical Finance 5, 263–279.

Donaldson, R.G., Kim, H.Y., 1993. Price barriers in the Dow Jones Industrial Average. Journal of Financial and Quantitative Analysis 28, 313–330.

Dorfleitner, G., Klein, C., 2009. Psychological barriers in European stock markets: Where are they? Global Finance Journal 19, 268–285.

Engelberg, J.E., Parsons, C.A., 2011. The causal impact of media in financial markets. Journal of Finance 66, 67–97.

George, T.J., Hwang, C.Y., 2004. The 52-week high and momentum profits. Journal of Finance 59, 2145–2176.

Harris, L., 1991. Stock price clustering and discreteness. Review of Financial Studies 4, 389–415.

Hirshleifer, D., Teoh, S.H., 2003. Limited attention, information disclosure, and financial reporting. Journal of Accounting and Economics 36, 337–386.

Hong, H., Stein, J.C., 1999. A unified theory of underreaction, momentum trading, and overreaction in asset markets. Journal of Finance 54, 2143–2184.

Hou, K., Moskowitz, T.J., 2005. Market frictions, price delay, and the cross-section of expected returns. Review of Financial Studies 18, 981–1020.

Imai, K., Keele, L., Tingley, D., 2010. A general approach to causal mediation analysis. Psychological Methods 15, 309–334.

Kacperczyk, M., Van Nieuwerburgh, S., Veldkamp, L., 2016. A rational theory of mutual funds' attention allocation. Econometrica 84, 571–626.

Koedijk, K.G., Stork, P.A., 1994. Should we care? Psychological barriers in stock markets. Economics Letters 44, 427–432.

Kolari, J.W., Pynnonen, S., 2010. Event study testing with cross-sectional correlation of abnormal returns. Review of Financial Studies 23, 3996–4025.

Kuo, W.Y., Lin, T.C., Zhao, J., 2015. Cognitive limitation and investment performance: Evidence from limit order clustering. Review of Financial Studies 28, 838–875.

Lacetera, N., Pope, D.G., Sydnor, J.R., 2012. Heuristic thinking and limited attention in the car market. American Economic Review 102, 2206–2236.

Ley, E., Varian, H.R., 1994. Are there psychological barriers in the Dow-Jones index? Applied Financial Economics 4, 217–224.

MacKinnon, J.G., Webb, M.D., 2017. Wild bootstrap inference for wildly different cluster sizes. Journal of Applied Econometrics 32, 233–254.

Mehta, J., Starmer, C., Sugden, R., 1994. Focal points in pure coordination games: An experimental investigation. Theory and Decision 36, 163–185.

Niederhoffer, V., 1965. Clustering of stock prices. Operations Research 13, 258–265.

Osler, C.L., 2003. Currency orders and exchange rate dynamics: An explanation for the predictive success of technical analysis. Journal of Finance 58, 1791–1819.

Patell, J.M., 1976. Corporate forecasts of earnings per share and stock price behavior: Empirical tests. Journal of Accounting Research 14, 246–276.

Peng, L., Xiong, W., 2006. Investor attention, overconfidence and category learning. Journal of Financial Economics 80, 563–602.

Peress, J., 2014. The media and the diffusion of information in financial markets: Evidence from newspaper strikes. Journal of Finance 69, 2007–2043.

Schelling, T.C., 1960. The Strategy of Conflict. Harvard University Press, Cambridge, MA.

Sicherman, N., Loewenstein, G., Seppi, D.J., Utkus, S.P., 2016. Financial attention. Review of Financial Studies 29, 863–897.

Tetlock, P.C., 2007. Giving content to investor sentiment: The role of media in the stock market. Journal of Finance 62, 1139–1168.

Webb, M.D., 2023. Reworking wild bootstrap-based inference for clustered errors. Canadian Journal of Economics 56, 839–858.

Westerhoff, F., 2003. Anchoring and psychological barriers in foreign exchange markets. Journal of Behavioral Finance 4, 65–70.

Yuan, Y., 2015. Market-wide attention, trading, and stock returns. Journal of Financial Economics 116, 548–564.

Zhang, X., 2024. Can daily closing prices predict future movements? The role of limit order clustering. Working Paper, University of Maryland, SSRN 4718961.